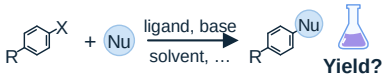
Chemical reaction optimization



Discrete choices dominate the search space

Ligands	 PPh$_3$	 PCy$_3$	 XPhos	…
Bases	K_2CO_3	Cs_2CO_3	KO*t*Bu	…
Solvents	 THF	 DMF	 DMSO	…

Huge discrete spaces, hours per experiment

LLMs know chemistry, but…



What is the yield with XPhos and Cs_2CO_3 in THF?

I think it will be 78%.

✗ ρ = 0.23 with measured yields

Rate the basicity of Cs_2CO_3 and of K_2CO_3 from 1 to 10.

Cs_2CO_3: 7, K_2CO_3: 7

✗ 66% agree with descriptor order

Hard to calibrate	Sensitive to prompting	Costly to query

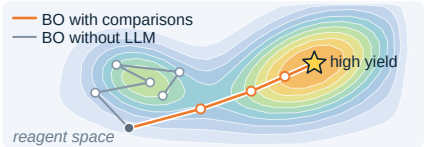
Comparative questions are more reliable



Which is more basic: Cs_2CO_3 or K_2CO_3?

Cs_2CO_3

✓ 82% agree with descriptor order



No calibration needed	Consistent across prompts	Cheap to query